

Comparison principles for stochastic heat equation

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Stochastic Webinar

Beijing Institute of Technology
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Outline

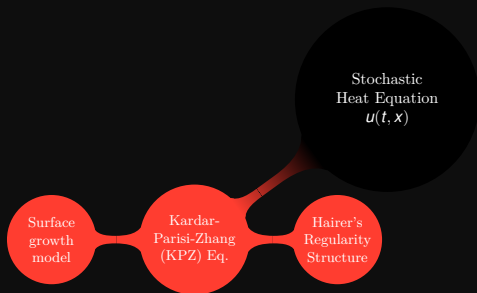
Introduction to SHE

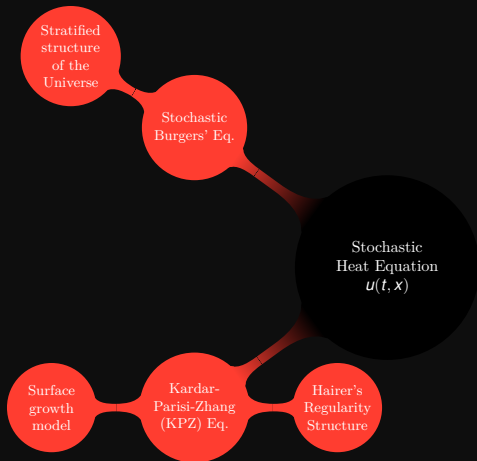
Comparison principles: Overview

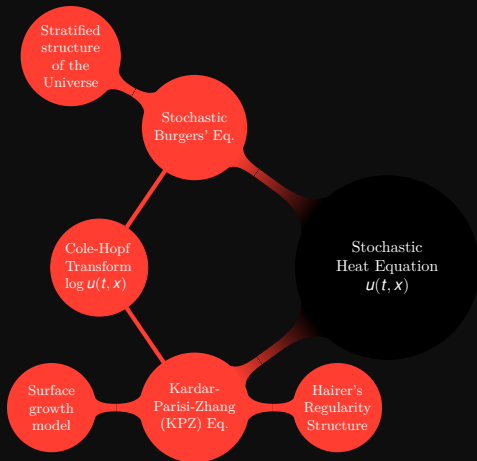
Stochastic comparison principles

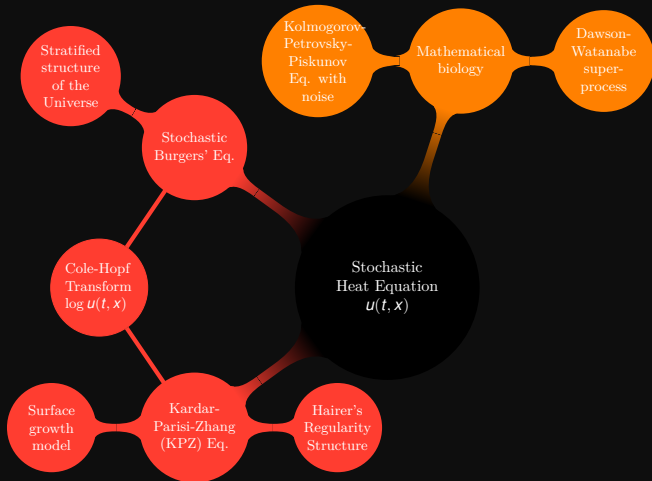
Sample-path comparison principles

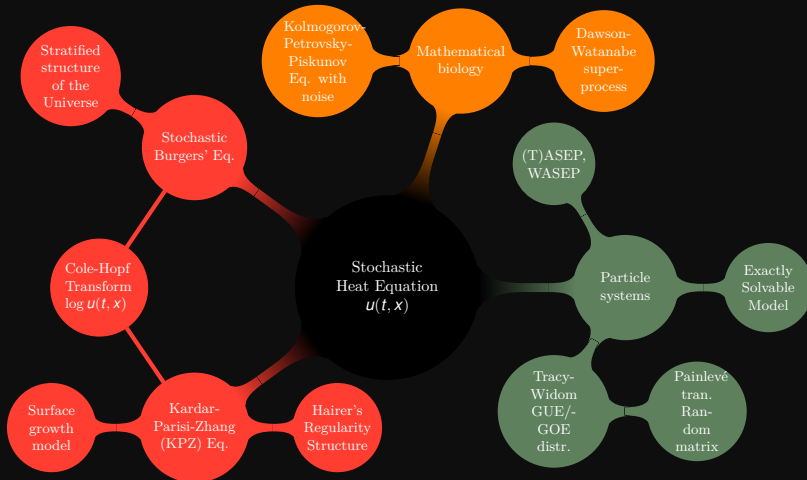
Stochastic
Heat Equation
 $u(t, x)$





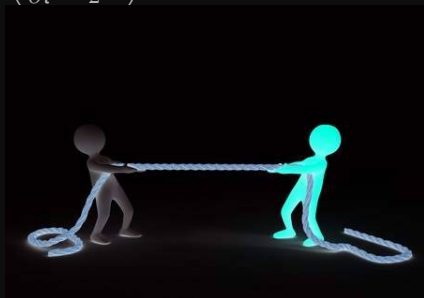








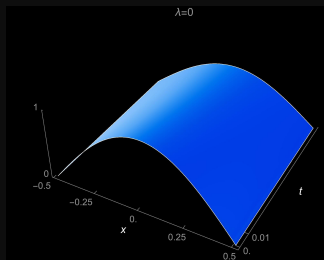
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 Δ
 $\rho(u) \dot{W}$

Smoothing

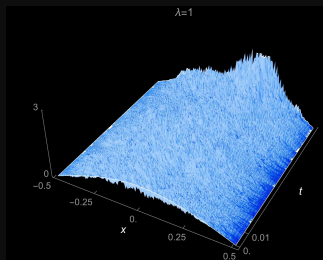
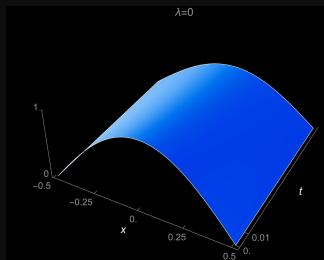
Roughening

$$\left(\frac{\partial}{\partial t} - \frac{\partial^2}{\partial x^2} \right) u(t, x) = \lambda u(t, x) \dot{W}(t, x), \quad x \in [-1/2, 1/2] \text{ with } u(0, x) = \cos(\pi x)$$



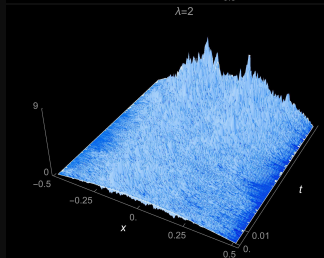
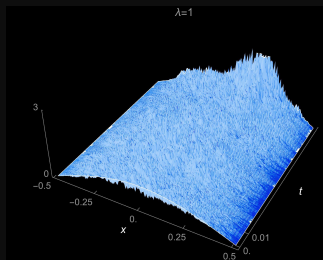
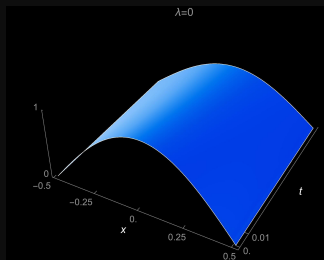
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Bearded SHE



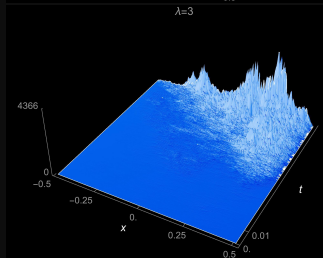
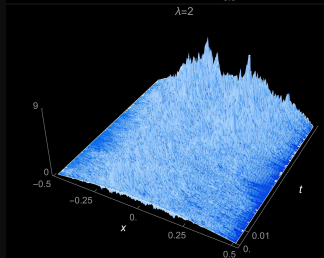
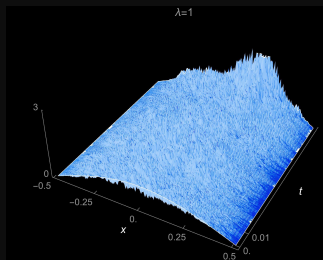
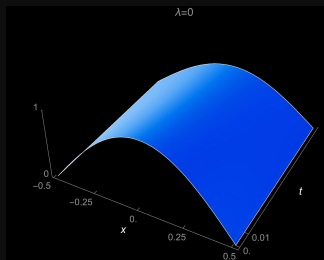
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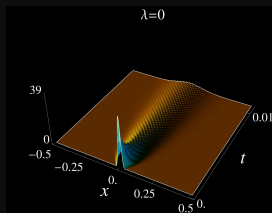
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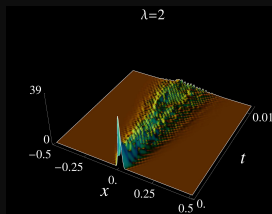
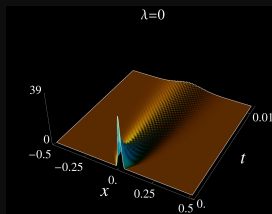
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with localized initial data



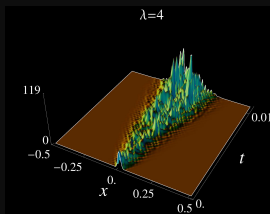
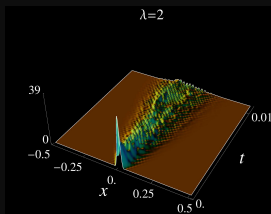
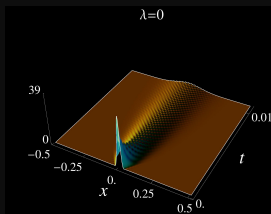
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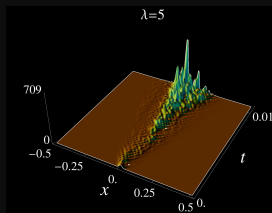
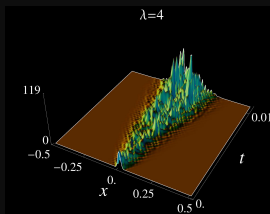
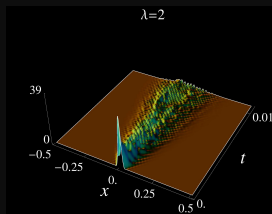
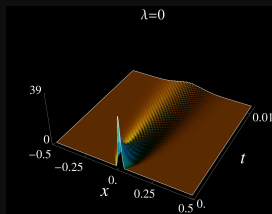
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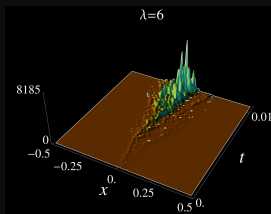
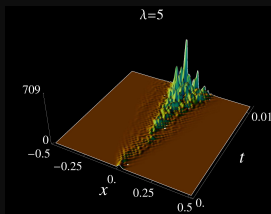
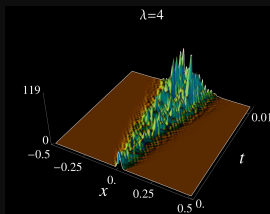
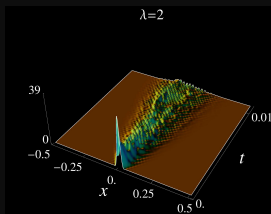
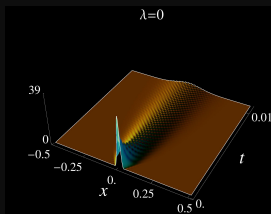
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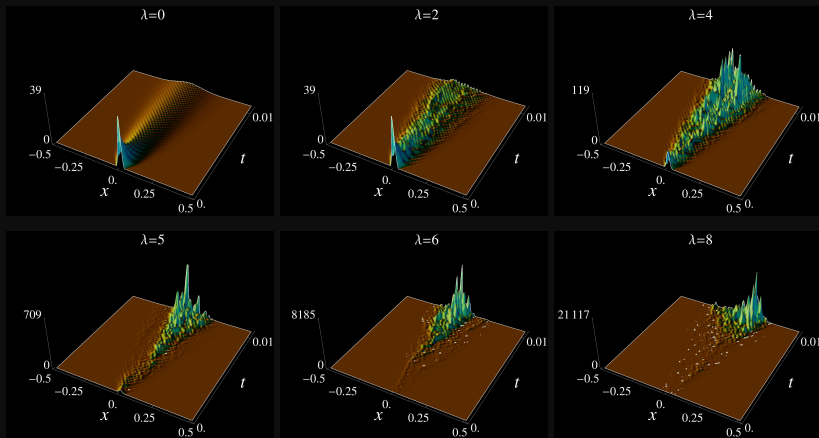
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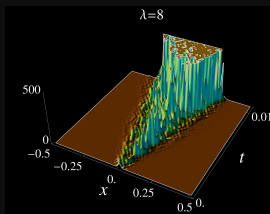
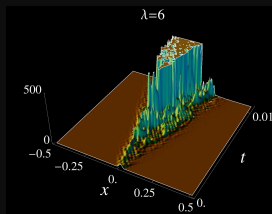
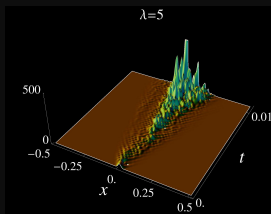
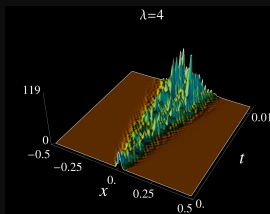
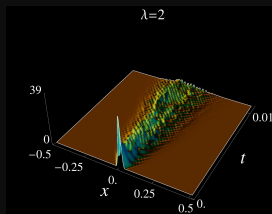
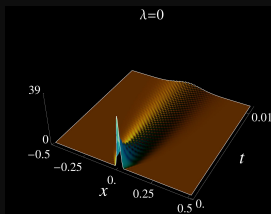
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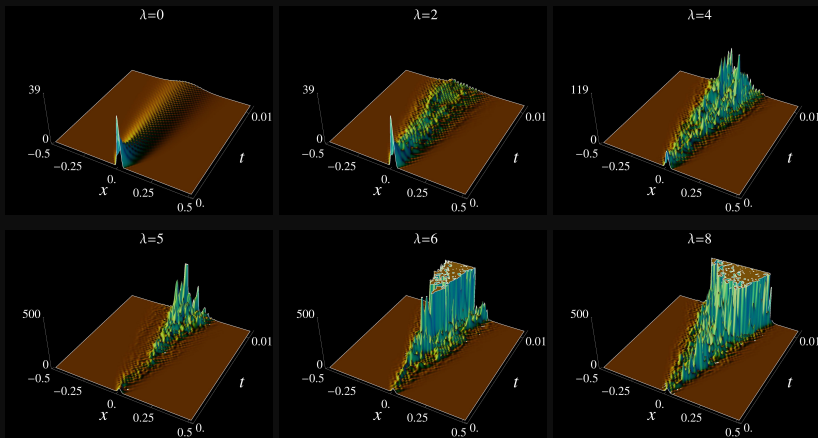
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The rate of the propagation of the tall peaks $\asymp \lambda^2$

[L.C. & Dalang, 15].

Various approaches for SPDE's

► – Semigroup approach (Da Prato, et al).

$$u(t) \in L^p(\Omega, \mathcal{H})$$

– Variational approach (Röckner, et al).

Gel'fand triple

► Random field approach (Walsh, Dalang, et al.)

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► – Regularity structure (Hairer):

Singular SPDEs

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The approach that we are using!

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Stochastic Heat Equation on $\mathbb{R}^d$

$$\begin{cases} \left(\frac{\partial}{\partial t} - \frac{1}{2} \Delta \right) u(t, x) = \rho(u(t, x)) \dot{W}(t, x), & t > 0, x \in \mathbb{R}^d \\ u(0, \cdot) = \mu(\cdot) \end{cases} \quad (\text{SHE})$$

1. ρ is globally Lipschitz continuous s.t. $\rho(0) = 0$.
2. μ is *rough* (measure+integrability condition).
3. $\dot{W}$ is white in time and colored in space (f nonneg. & nonneg. definite):

$$\mathbb{E} \left[\dot{W}(t, x) \dot{W}(s, y) \right] = \delta_0(t - s) f(x - y)$$

4. $\rho(u) = \lambda u$: *Parabolic Anderson Model*.

Mild solution

$$u(t, x) = J_0(t, x) + \underbrace{\int_0^t \int_{\mathbb{R}^d} G(t-s, x-y) \rho(u(s, y)) W(\mathrm{d}s \mathrm{d}y)}_{=: I(t, x)}$$

$$\dagger \quad G(t, x) = (2\pi t)^{-d/2} \exp\left(-\frac{|x|^2}{2t}\right)$$

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Comparison principles for SHE

$$\begin{cases} \left(\frac{\partial}{\partial t} - \frac{1}{2} \Delta \right) u(t, x) = \rho(u(t, x)) \dot{W}(t, x), & t > 0, x \in \mathbb{R}^d \\ u(0, \cdot) = \mu(\cdot) \end{cases} \quad (\text{SHE})$$

Let u_1 and u_2 be two solutions to (SHE) obtained either by

$$\mu \rightarrow \mu_1, \mu_2 \quad \text{or} \quad \rho \rightarrow \rho_1, \rho_2 \quad \text{or} \quad \dot{W} \rightarrow \dot{W}_1, \dot{W}_2.$$

Question Can we compare u_1 & u_2 ?

In which sense?

Why is it important?

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Sample-path comparison	Stochastic (moment) comparison
$\mu_1 > \mu_2$ $\Downarrow$ $u_1(t, x) > u_2(t, x) \quad \forall (t, x) \text{ a.s.}$	$\rho_1(\cdot) \geq \rho_2(\cdot) \quad \text{or} \quad f_1 \geq f_2$ $\Downarrow$ $\mathbb{E}(u_1(t, x)^k) \geq \mathbb{E}(u_2(t, x)^k) \quad \forall (t, x), k \geq 1$
Strict positivity $\sim$ Hopf-Cole Sol. Neg. moments $\sim$ Smooth density	No Feynman-Kac formulas for nonlinear ρ Moment growth in $k \sim$ intermittency
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Plan

Introduction to SHE

Comparison principles: Overview

Stochastic comparison principles

Sample-path comparison principles

Examples of stochastic comparisons

Assume $\mu_1 = \mu_2$ and $\dot{W}_1 = \dot{W}_2$ but ρ 's are different:

either $\rho_1(x) \leq \rho_2(x) \leq 0$ and $\rho_1(x) \geq \rho_2(x) \geq 0$ for all $x \geq 0$,

or

Assume $\mu_1 = \mu_2$ and $\rho_1 = \rho_2$, but $\dot{W}$'s are different: $\dot{f}_1 \geq \dot{f}_2$.

Eg. 1 *Moment comparison principle*: $\forall m \in \mathbb{N}$, $\forall (t_\ell, x_\ell) \in (0, \infty) \times \mathbb{R}^d$, $\forall k_\ell \in \mathbb{N}$, $\ell = 1, \dots, m$,

$$\mathbb{E} \left[\prod_{\ell=1}^m U_1^{k_\ell}(t_\ell, x_\ell) \right] \geq \mathbb{E} \left[\prod_{\ell=1}^m U_2^{k_\ell}(t_\ell, x_\ell) \right].$$

(This is the content in [Joseph, et al '17] and [Foondun, et al '18])

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$$\mathbb{E} \left[\prod_{\ell=1}^m u_1^{k_\ell}(t_\ell, x_\ell) \right] \geq \mathbb{E} \left[\prod_{\ell=1}^m u_2^{k_\ell}(t_\ell, x_\ell) \right].$$

(This is the content in [Joseph, et al '17] and [Foondun, et al '18])

Examples of stochastic comparisons

Assume $\mu_1 = \mu_2$ and $\dot{W}_1 = \dot{W}_2$ but ρ 's are different:

either $\rho_1(x) \leq \rho_2(x) \leq 0$ and $\rho_1(x) \geq \rho_2(x) \geq 0$ for all $x \geq 0$,

or

Assume $\mu_1 = \mu_2$ and $\rho_1 = \rho_2$, but $\dot{W}$'s are different: $f_1 \geq f_2$.

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Eg. 2 $\forall m \in \mathbb{N}, \forall (t_\ell, \mathbf{x}_\ell) \in (0, \infty) \times \mathbb{R}^d, \forall k_\ell \in \mathbb{N}, \ell = 1, \dots, m,$

$$\mathbb{E} \left[\prod_{\ell=1}^m g_\ell^{k_\ell}(u_1(t_\ell, \mathbf{x}_\ell)) \right] \geq \mathbb{E} \left[\prod_{\ell=1}^m g_\ell^{k_\ell}(u_2(t_\ell, \mathbf{x}_\ell)) \right], \quad (\star)$$

where $g_\ell(z)$ can be either

$$\mathbf{x}^{b_\ell} [\log(\mathbf{c}_\ell + \mathbf{x})]^{a_\ell} \quad \text{or} \quad \mathbf{x}^{d_\ell},$$

with $a_\ell, b_\ell, d_\ell \geq 1$ and $c_\ell \geq e$.

Eg. 3 Statement in $(\star)$ is true with $g_\ell(z)$ being either

$$\exp(-\lambda_\ell z), \quad \frac{1}{(1+z)^{c_\ell}}, \quad \text{or} \quad \log\left(\frac{z+a_\ell}{z+b_\ell}\right)$$

with $\lambda_\ell > 0, a_\ell > b_\ell > 0$ and $c_\ell \geq 1$.

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Eg. 4 $\forall m \geq 1, t_m > \cdots > t_1 > 0, k_1, \cdots, k_m \in \mathbb{N} \setminus \{0\}, \alpha_1, \cdots, \alpha_m \in [2, \infty)$, and $\mathbf{x}_j^\ell \in \mathbb{R}^d$ with $\ell = 1, \cdots, m$ and $j = 1, \cdots, k_\ell$ s.t. $\{\mathbf{x}_{k_1}^\ell, \cdots, \mathbf{x}_{k_\ell}^\ell\}$ are distinct points for each ℓ ,

$$\begin{aligned} \mathbb{E} \left(\prod_{\ell=1}^m \left[u_1^2(t_\ell, \mathbf{x}_1^\ell) + \cdots + u_1^2(t_\ell, \mathbf{x}_{k_\ell}^\ell) \right]^{\frac{\alpha_\ell}{2}} \right) \\ \geq \mathbb{E} \left(\prod_{\ell=1}^m \left[u_2^2(t_\ell, \mathbf{x}_1^\ell) + \cdots + u_2^2(t_\ell, \mathbf{x}_{k_\ell}^\ell) \right]^{\frac{\alpha_\ell}{2}} \right). \end{aligned}$$

Eg. 5 $\forall (t, x) \in (0, \infty) \times \mathbb{R}^d$, $c > 0$ and any integer $n \geq 1$, it holds that

$$\mathbb{E} \left([u_1(t, x) - c]^{2n} \right) \geq \mathbb{E} \left([u_2(t, x) - c]^{2n} \right). \quad (\star)$$

In particular, by choosing $c = (\mu * G(t, \cdot))(x)$, $(\star)$ tells us that all *centered moments* of even orders can be compared, i.e.,

$$\mathbb{E} \left(h_1(t, x)^{2n} \right) \geq \mathbb{E} \left(h_2(t, x)^{2n} \right).$$

† Recall: $u_\ell(t, x) = J_0(t, x) + h_\ell(t, x)$, $\ell = 1, 2$.

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Eg. 6 (*Slepian's inequality for SPDEs*)

If $\mu_1 = \mu_2$ and $\rho_1 = \rho_2$, and if $f_1 \geq f_2$ but f_1 and f_2 are *equal* to each other near the origin, then for all $a > 0$, $t > 0$, and $x_1, \dots, x_N \in \mathbb{R}^d$,

$$\mathbb{P} \left\{ \max_{1 \leq k \leq N} u_1(t, x_k) \leq a \right\} \geq \mathbb{P} \left\{ \max_{1 \leq k \leq N} u_2(t, x_k) \leq a \right\}.$$

Formally, this is equivalent to take $g(x) = \mathbb{I}_{[0,a]}(x)$.

Difficulties and hence contributions

1. **Dalang's condition** — Weakest assumption for the correlation measure to have $L^2(\Omega)$ solution:

$$\int_{\mathbb{R}^d} \frac{\widehat{f}(d\xi)}{1 + |\xi|^2} < \infty. \quad (\text{DL})$$

2. **Rough initial data:** μ is a (nonnegative) measure s.t.

$$J_0(t, x) < \infty \quad \forall t > 0, x \in \mathbb{R}^d \quad \iff \quad \int_{\mathbb{R}^d} e^{-|x|^2/(4t)} \mu(dx) < \infty, \forall t > 0$$

$$\text{but } \int_{\mathbb{R}^d} e^{-|x|^2/(4t)} \mu(dx) = e^{-|x|^2/(4t)} \mu(dx) \rightarrow \infty, \text{ as } t \rightarrow 0, \forall x \in \mathbb{R}^d$$

3. We have to go beyond the moment condition and covers, e.g., examples 2-4.

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3. **Nonlinear problems** — How to extend the previous condition and the previous results to nonlinear problems?

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3. **Nonlinearities** — How to deal with the nonlinear correlation and the nonlinear complex-valued functions?

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$$J_0(t, x) = \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} e^{-a|x-y|^2} \mu(dy) \mu(dx) \quad (x, y \in \mathbb{R}^d, t \in \mathbb{R}_+)$$

3. **Nonlocality of the problem** — the nonlocal correlation and the rough initial data complicate the problem.

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$$\text{E.g.,} \quad \mu = \delta_0, \quad \mu(\mathrm{d}x) = e^{+|x|^{3/2}} \mathrm{d}x, \quad \mu(\mathrm{d}x) = |x|^{-\beta} \mathrm{d}x \ (\beta \in (0, d)), \dots$$

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Multiple-time comparison over certain function sets $\mathbb{F}$

Def. Let $\{u_i(t, x); (t, x) \in \mathbb{R}_+ \times K\}$, $i = 1, 2$, be two random fields, where K is the spatial index set.

For some function cone $\mathbb{F}$, and some $n \geq 1$, we say that u_1 and u_2 satisfy

*the multiple/n-time stochastic comparison principle
over $\mathbb{F}$ with u_1 dominating u_2*

if for any $0 < t_1 < \cdots < t_n < \infty$, and $F_1, \dots, F_n \in \mathbb{F}$, it holds that

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Statement of main results

Theorem. (L.C. & Kim '20) Under Dalang's condition, for rough initial data,

$$\text{if } \rho_1(\cdot) \geq \rho_2(\cdot) \geq 0 \quad \text{or} \quad \text{if } f_1 \geq f_2,$$

then for $K = \mathbb{R}^d$, u_1 and u_2 satisfy

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2. the multiple-time stochastic comparison principle over $\mathcal{C}_b^1$,
3. the 1-time stochastic comparison principle over $\mathcal{C}_b^1$.

ρ : polynomial growth. $+$: first derivative nonnegative.

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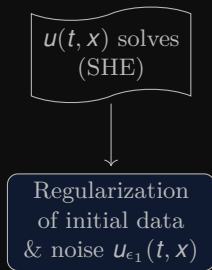
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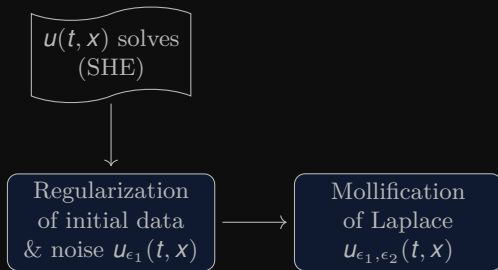
Proof strategy: approximations of SHE

$u(t, x)$ solves
(SHE)

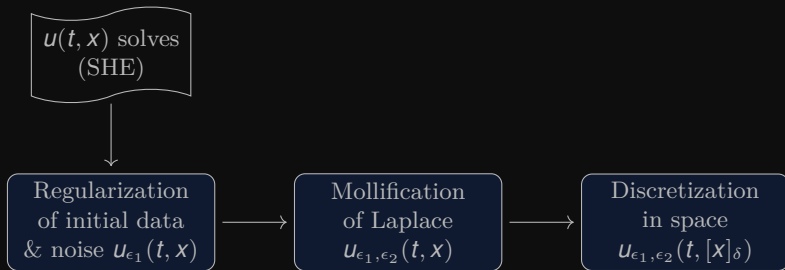
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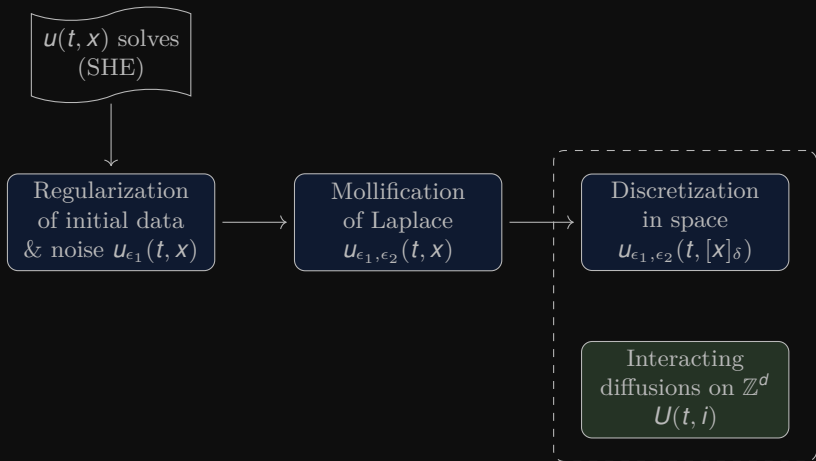
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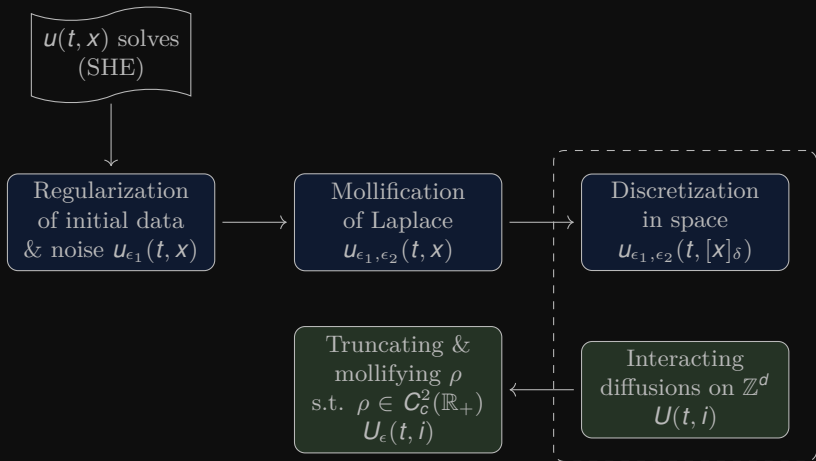
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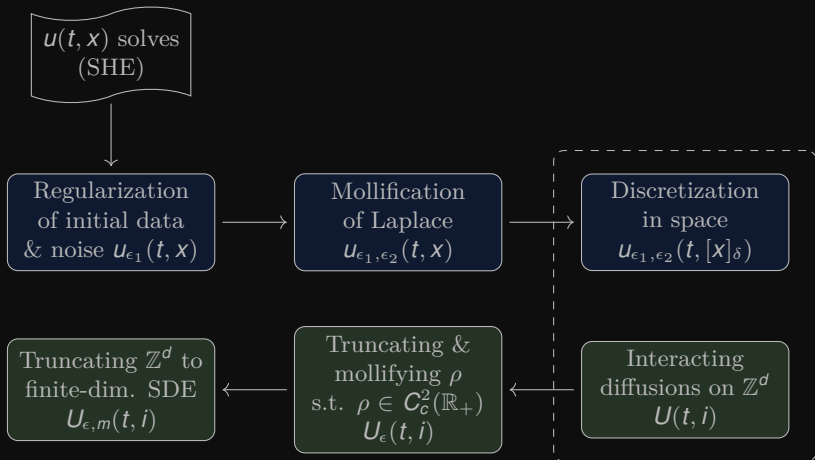
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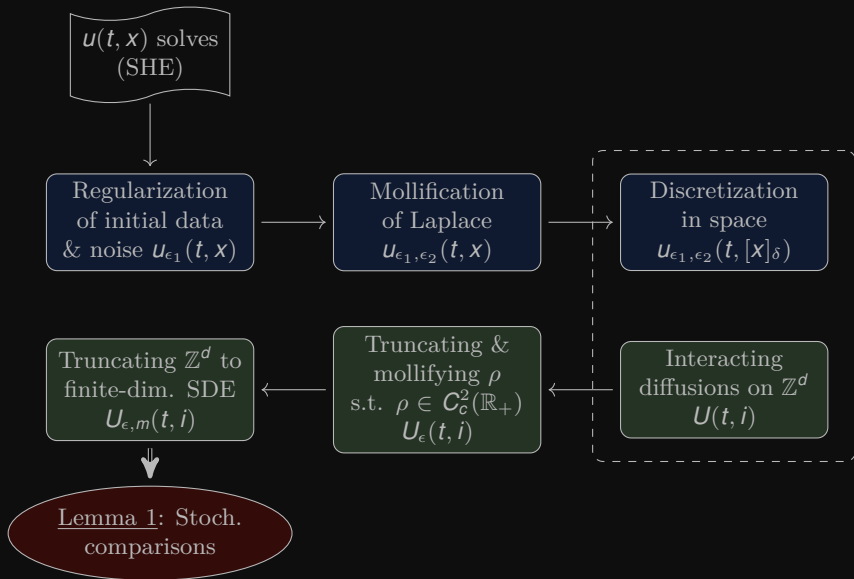
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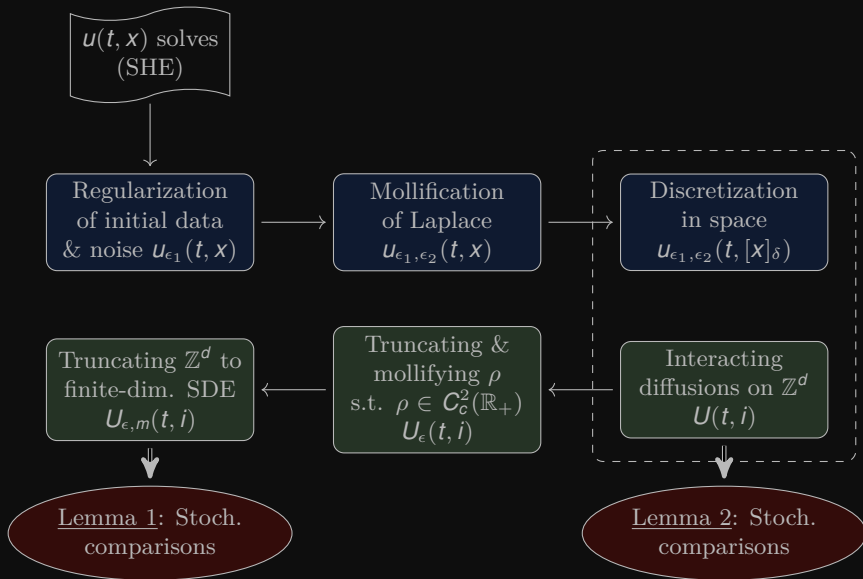
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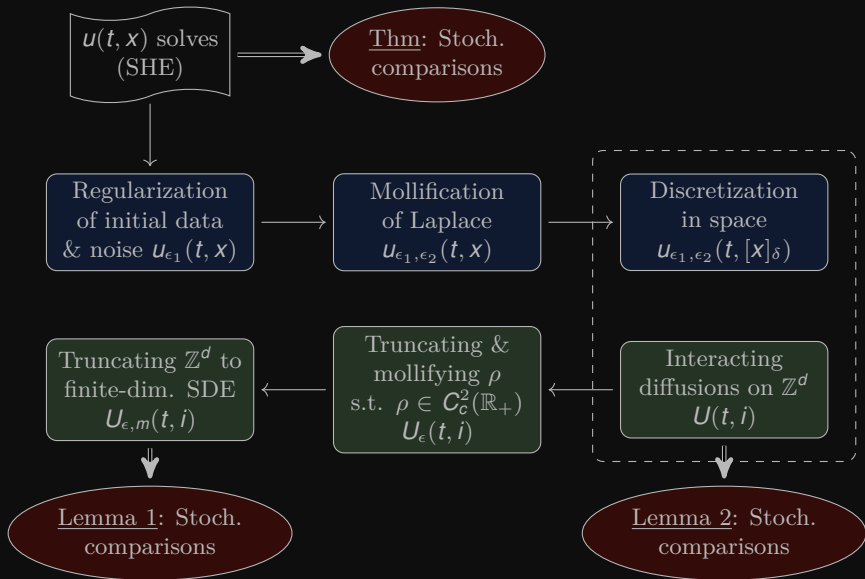
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$$dU(t, i) = \kappa \sum_{j \in K} p_{i,j} (U(t, j) - U(t, i)) dt + \rho(U(t, i)) dM_i(t), \quad i \in K$$

Lemma 1. (L.C. & Kim '20) For the finite (i.e., $|K| < \infty$) interacting diffusion eq.,

$$\text{if } \rho_1(\cdot) \geq \rho_2(\cdot) \geq 0 \quad \text{or} \quad \text{if } \gamma_1 \geq \gamma_2,$$

where $\rho_i \in \mathcal{C}_c^2(\mathbb{R}_+)$, then u_1 and u_2 satisfy all the comparison results in Theorem.

Ideas of proof: Follows closely to [Cox, et al '96]:

Integration-by-parts formula

Trotter's product formula

Markov properties (for n -time case)

with necessary changes to make in order to take care of

- ▶ correlated noise and
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Let $T_t^{(\ell)}$ be the semigroup, i.e., $T_t^{(\ell)} F(x) := \mathbb{E}_x [F(U_\ell(t, \cdot))]$ and G_ℓ be the corresponding infinitesimal generator, $\ell \in \{1, 2\}$:

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► **Integration-by-parts formula**

$$T_t^{(1)} - T_t^{(2)} = \int_0^t T_s^{(1)} \left[G^{(1)} - G^{(2)} \right] T_{t-s}^{(2)} ds$$

► We only need to show that

$$D_i D_j T_t F(z) \geq 0 \quad \forall i, j, t, z.$$

$$\underbrace{\kappa \sum_{i,j} (D_i D_j F)(z)}_{\geq 0} + \underbrace{\sum_{i,j} (D_i D_j F)(z)}_{\geq 0}$$

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$$\underbrace{(D_i D_j F)}_{\geq 0} \underbrace{TF_t(z)}_{\geq 0}$$

Let $T_t^{(\ell)}$ be the semigroup, i.e., $T_t^{(\ell)} F(x) := \mathbb{E}_x [F(U_\ell(t, \cdot))]$ and G_ℓ be the corresponding infinitesimal generator, $\ell \in \{1, 2\}$:

$$G^{(\ell)} = \kappa \sum_{1 \leq i, j \leq d} (p_{i,j} - \delta_{i,j}) z_j D_i + \begin{cases} \frac{1}{2} \sum_{1 \leq i, j \leq d} \rho_\ell(x_i) \rho_\ell(x_j) \gamma(i-j) D_i D_j \\ \frac{1}{2} \sum_{1 \leq i, j \leq d} \rho(x_i) \rho(x_j) \gamma_\ell(i-j) D_i D_j \end{cases}$$

► **Integration-by-parts formula**

$$T_t^{(1)} - T_t^{(2)} = \int_0^t T_s^{(1)} \left[G^{(1)} - G^{(2)} \right] T_{t-s}^{(2)} ds$$

► We only need to show that

$$\begin{aligned} D_i D_j T_t F(z) &\geq 0 \quad \forall i, j, t, z. \\ &\parallel \\ \underbrace{(D_i D_j F)}_{\geq 0} &\underbrace{TF_t(z)}_{\stackrel{?}{\geq 0}} \end{aligned}$$

$$\begin{array}{ccc}
& G^{(\kappa, \rho)} & \\
& || & \\
\underbrace{\kappa \sum_{1 \leq i, j \leq d} (p_{i,j} - \delta_{i,j}) z_j D_i}_{:= G^{(\kappa, 0)}} & + & \underbrace{\frac{1}{2} \sum_{1 \leq i, j \leq d} \rho(x_i) \rho(x_j) \gamma(i-j) D_i D_j}_{:= G^{(0, \rho)}}
\end{array}$$

$$\begin{cases} dU(t, i) = \rho(U(t, i)) dM_i(t) \\ U(0, i) = u_0(i) \end{cases}$$

$$\begin{array}{c} \wr \\ T^{(\kappa, 0)} \end{array}$$

$$\begin{cases} dU(t, i) = \kappa \sum_{j=1}^d (p_{i,j} - \delta_{i,j}) U(s, j) dt \\ U(0, i) = u_0(i) \end{cases}$$

$$\begin{array}{c} \wr \\ T^{(0, \rho)} \end{array}$$

Trotter's product formula

$$\lim_{k \rightarrow \infty} \left[T_{t/k}^{(\kappa, 0)} T_{t/k}^{(0, \rho)} \right]^k = T_t^{(\kappa, \rho)}$$

□

$$dU(t, i) = \kappa \sum_{j \in \mathbb{Z}^d} p_{i,j} (U(t, j) - U(t, i)) dt + \rho(U(t, i)) dM_i(t), \quad i \in \mathbb{Z}^d$$

Lemma 2. (L.C. & Kim '20) For the infinite interacting diffusion eq.,

$$\text{if } \rho_1(\cdot) \geq \rho_2(\cdot) \geq 0 \quad \text{or} \quad \text{if } \gamma_1 \geq \gamma_2,$$

where $\rho_i \in \text{Lip}$, then u_1 and u_2 satisfy all the comparison results in Theorem.

	Cox, et al PTRF '96	Foondun, et al AAP '18	L.C. & Kim EJP '20
Noise	i.i.d. BM's	Correlated BM's	
Fun. Cones $\mathbb{F}$	$\mathbb{F}[C_b^{2,\nu}]$	$x^n, n \in \mathbb{N}$ (moment comparison)	$\mathbb{F}[C_p^{2,\nu}]$; Slepian's ineq.
Comparing	only ρ		both ρ and γ

$$dU(t, i) = \kappa \sum_{j \in \mathbb{Z}^d} p_{i,j} (U(t, j) - U(t, i)) dt + \rho(U(t, i)) dM_i(t), \quad i \in \mathbb{Z}^d$$

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Noise	i.i.d. BM's	Correlated BM's	
Fun. Cones $\mathbb{F}$	$\mathbb{F}[\mathcal{C}_b^{2,\nu}]$	$x^n, n \in \mathbb{N}$ (moment comparison)	$\mathbb{F}[\mathcal{C}_\rho^{2,\nu}]$; Slepian's ineq.
Comparing	only ρ		both ρ and γ

Theorem. (L.C. & Kim '20) Under Dalang's condition, for rough initial data,

$$\text{if } \rho_1(\cdot) \geq \rho_2(\cdot) \geq 0 \quad \text{or} \quad \text{if } f_1 \geq f_2,$$

then for $K = \mathbb{R}^d$, u_1 and u_2 , solutions to (SHE), satisfy

1. the multiple-time stochastic comparison principle over $\mathbb{F}[\mathcal{C}_{\rho,+}^{2,\nu}]$,
2. the multiple-time stochastic comparison principle over $\mathbb{F}[\mathcal{C}_{b,-}^{2,\nu}]$,
3. the 1-time stochastic comparison principle over $\mathbb{F}[\mathcal{C}_{\rho}^{2,\nu}]$.

ρ : polynomial growth. $+$: first derivative nonnegative.

b : bounded. $-$: first derivative nonpositive.

ν : convex functions.

	Joseph, et al AOP '17	Foondun, et al AAP '18	L.C. & Kim EJP '20
Equation	SHE on $\mathbb{R}$	Frac. SHE on $\mathbb{R}^d$	SHE on $\mathbb{R}^d$
Noise	$f = \delta_0$	Riesz kernel	Dalang's Condition
Initial data	$u_0(x) \equiv 1$	$u_0(x) \geq 0$ bounded	Rough
Function cones $\mathbb{F}$	$x^n, n \in \mathbb{N}$ (moment comparison)		More general; Slepian's ineq.
Comparing	only ρ		both ρ and $\dot{W}$
Mollification of Δ	Random walk		Yoshida approximation

Plan

Introduction to SHE

Comparison principles: Overview

Stochastic comparison principles

Sample-path comparison principles

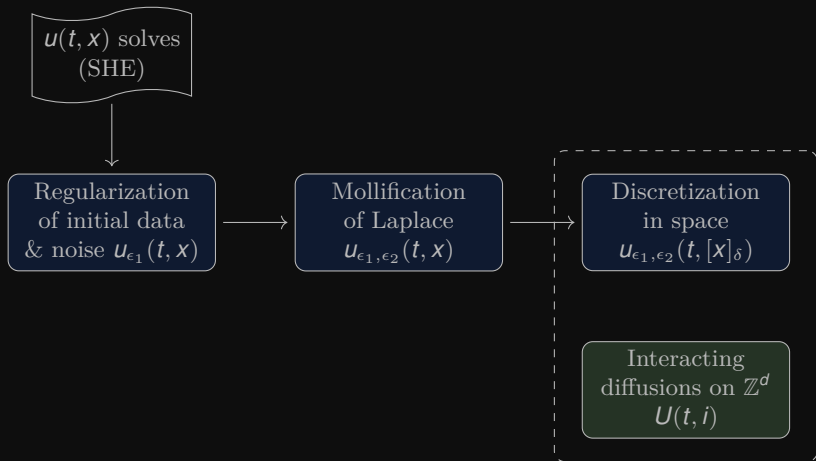
Sample-path comparison	Stochastic (moment) comparison
$\mu_1 > \mu_2$ $\Downarrow$ $u_1(t, x) > u_2(t, x) \quad \forall(t, x) \text{ a.s.}$	$\rho_1(\cdot) \geq \rho_2(\cdot) \quad \text{or} \quad f_1 \geq f_2$ $\Downarrow$ $\mathbb{E}(u_1(t, x)^k) \geq \mathbb{E}(u_2(t, x)^k) \quad \forall(t, x), k \geq 1$
Strict positivity $\sim$ Hopf-Cole Sol. Neg. moments $\sim$ Smooth density	No Feynman-Kac formulas for nonlinear ρ Moment growth in $k \sim$ intermittency
Mueller '91, Shiga '94 Tessitore & Zabczyk '98 L.C. & Kim '15, L.C. & Huang '18	Joseph, Khoshnevisan & Mueller '17 Foondun, Joseph & Li '18 L.C. & Kim '20

Two steps in the proof....

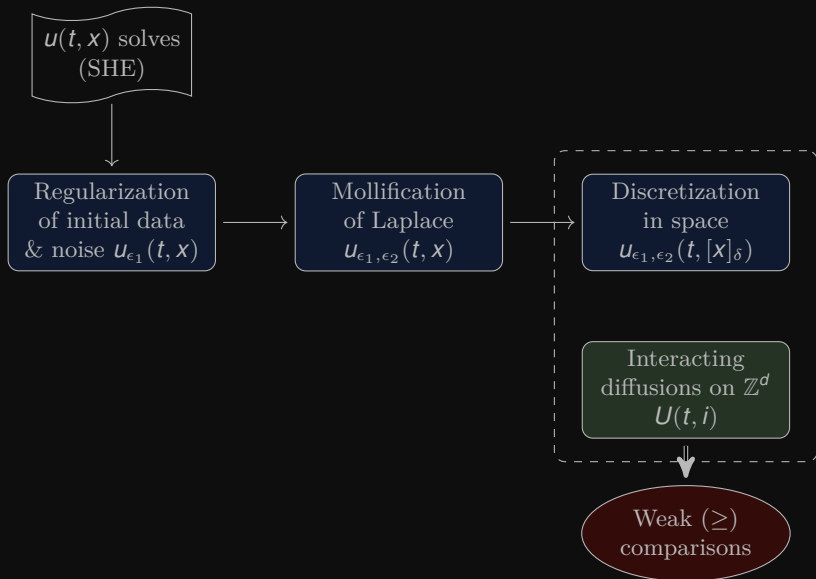
Step 1. Weak comparison: $u_1(t, x) \geq u_2(t, x)$ a.s.

Step 2. Strong comparison: $u_1(t, x) > u_2(t, x)$ a.s.

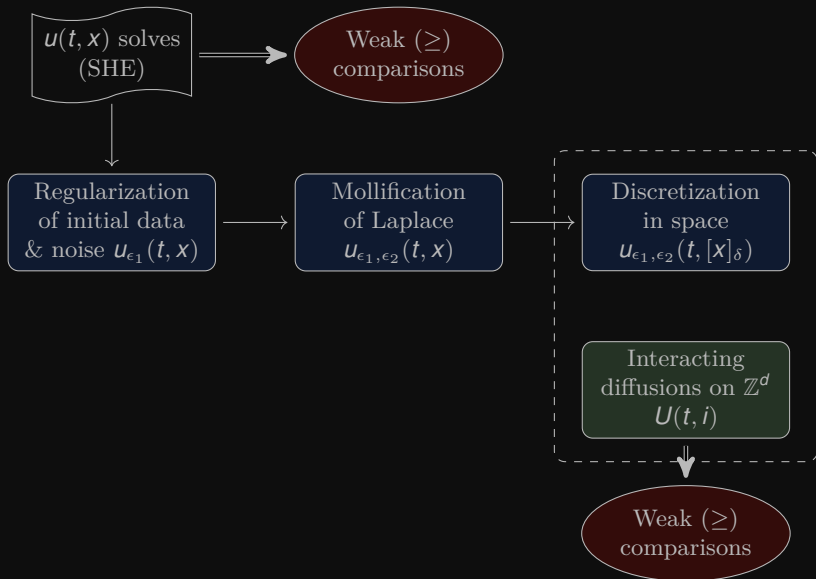
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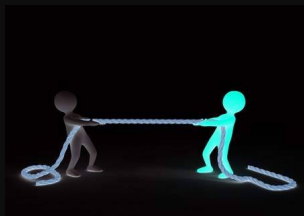
Weak comparisons: $u_1(t, x) \geq u_2(t, x)$ a.s.



Strong comparison: $u_1(t, x) > u_2(t, x)$ a.s.

Mueller's contribution

$$\begin{aligned} \mathbb{P} \left(u(s, x) = J_0(s, x) + I(s, x) < \beta \text{ for all } (s, x) \text{ in some times-space stripe } \mathcal{S} \right) \\ = \beta^{-p} \mathbb{E} \left(\sup_{(s, x) \in \mathcal{S}} |I(s, x)|^p \right) \leq \underbrace{C \exp \left(\frac{1}{4} \alpha p \log(\tau) + C p^{\frac{\alpha+1}{\alpha}} \tau \right)}_{=: f(p)} \end{aligned}$$

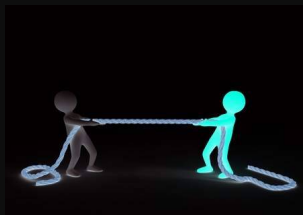


$$p = \left(\frac{\alpha^2 \log(1/\tau)}{4(\alpha+1)C\tau} \right)^\alpha$$

Strong comparison: $u_1(t, x) > u_2(t, x)$ a.s.

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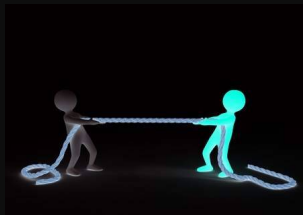


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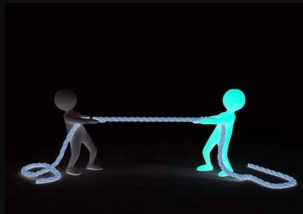


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	Mueller '91 Shiga '94	L.C. & Kim '05	Tessitore & Zabczyk '98	L.C. & Huang '19
Eq.	SHE on $\mathbb{R}$	Frac. SHE on $\mathbb{R}$	SHE on $\mathbb{R}^d$	
Noise	white in space		Colored; Strong restrictions on $\hat{f}$	Dalang's Condition+
Init. data	Functions	Rough	Functions	Rough

† Dalang' conditions+: $\alpha \in (0, 1]$,

$$\int_{\mathbb{R}^d} \frac{\widehat{f}(d\xi)}{(1 + |\xi|^2)^{1-\alpha}} < \infty. \quad (\text{DL+})$$

Sample-path comparison principle:

1. Mueller, C. On the support of solutions to the heat equation with noise. *Stochastics Stochastics Rep.* 1991
2. Shiga, T. Two contrasting properties of solutions for one-dimensional stochastic partial differential equations. *Canad. J. Math.* 1994
3. Tessitore, G. and J. Zabczyk. Strict positivity for stochastic heat equations. *Stochastic Process. Appl.* 1998
4. L.C. and K. Kim. On comparison principle and strict positivity of solutions to the nonlinear stochastic fractional heat equations. *Ann. Inst. Henri Poincaré Probab. Stat.* 2017
5. L.C. and J. Huang. Comparison principle for stochastic heat equation on $\mathbb{R}^d$. *Ann. Probab.* 2019

Stochastic (moment) comparison principle:

1. Cox, J. T., K. Fleischmann and A. Greven. Comparison of interacting diffusions and an application to their ergodic theory. *Probab. Theory Related Fields* 1996
2. Tribe, R. and J. Zabczyk. Notes on stochastic comparisons theorems for SDEs and SPDEs. *Unpublished note, private communication* 2013
3. Joseph, M., D. Khoshnevisan and C. Mueller. Strong invariance and noise-comparison principles for some parabolic stochastic PDEs. *Ann. Probab.* 2017
4. Foondun, M., M. Joseph and S.T. Li. An approximation result for a class of stochastic heat equations. *Ann. Appl. Probab.* 2018
5. L.C. and K. Kim. Stochastic comparison principle for stochastic heat equation. *Electron. J. Probab.* 2020



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- L.C. and K. Kim: On comparison principle and strict positivity of solutions to the nonlinear stochastic fractional heat equations. *Ann. Inst. Henri Poincaré Probab. Stat.* 53 (2017), no. 1, 358–388.
- L.C. and J. Huang: Comparison principle for stochastic heat equation on $\mathbb{R}^d$. *Ann. Probab.*, 2019, Vol. 47, No. 2, 989–1035.
- L.C. and K. Kim: Stochastic comparison for stochastic heat equation on $\mathbb{R}^d$. *Electron. J. Probab.*, 2020.

Thank you for listening!

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